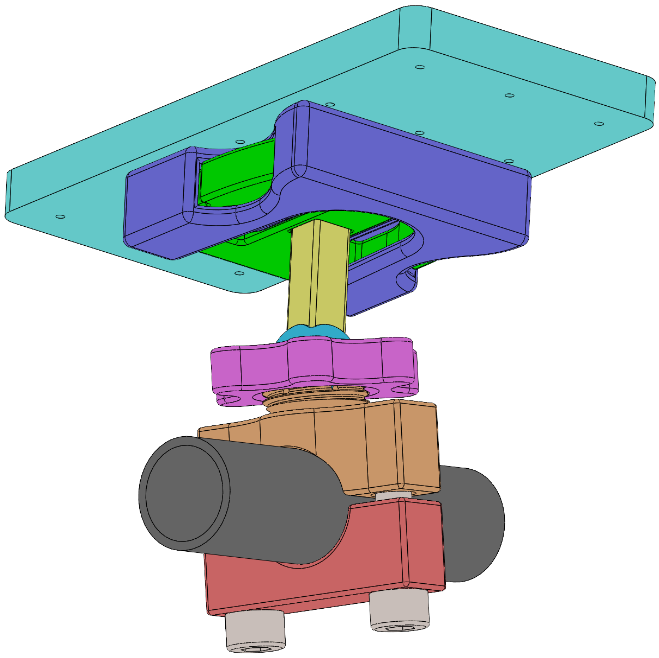
MEC4407

Concept Design Submission



Team Number: 47

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Executive Summary

This concept design report illustrates the initial development of ideas, preliminary CAD modelling and prototyping of a phone mount for motorbikes/bicycles. The product is being developed for a plastic goods manufacturing company with new Nylon 12 technology. This mount design needs to be universally adaptable and fit a phone of 150mm x 70mm x 10mm through a linear attachment motion with a rapid release. The mount is split into three main subsystems. The first is the clamp to the handlebar, parameterised for varying diameters but optimised for a 1 inch handlebar. The adjustment mechanism allows for pitch, roll and yaw to provide convenient viewing angles of the phone for the rider, and the tool free linear quick release phone attachment mechanism, with a part of it attached to the back of the phone.

To conceptualise ideas for practical design configurations, the team used an OFFERS analysis to map key functions, constraints and specifications. The team underwent a vigorous creative solution generation brainstorm before settling on specific design components to combine into the preliminary assembly. This initial product was prototyped and flaws were identified before being rectified, until a working physical model was produced. This model underwent Finite Element Analysis (FEA) in Ansys to ensure it maintains structural integrity under normal operating load cases aboard a motorbike.

The findings indicate that the current concept meets the functional requirements with minor deflections under a heavy bump load but holds strong under static testing. Moving forward into the next stage of design, we will look to improve the deflection, test the aerodynamic efficiency of the mount in order to further validate our design. Parts of the model will also be transitioned to using Nylon 12, as per the requirements of the project, since initial prototyping was done using PETG and PLA, with differing material characteristics.

Figure 1: Target User and Loading Condition for the Phone Mount: a Motorcyclist Cornering.

(https://cardosystems.com/blogs/cardo-blog/is-riding-a-motorcycle-hard)

Acknowledgement of Country

We acknowledge the Wurundjeri Woi Wurung people of the Kulin Nation, the Traditional Owners and Custodians of the land on which Monash University is located, and where all steps of this project have been undertaken. We pay our respects to their Elders past and present, and extend that respect to all Aboriginal and Torres Strait Islander peoples.

As engineering students, we recognise the importance of considering how the materials we select, products we design, and processes we use interact with the environment. In designing our phone mount, we acknowledge the responsibility to consider these factors and hence the potential environmental impact. We strive to respect and care for the environment in recognition of its enduring importance to the Wurundjeri Woi Wurung people of the Kulin Nation and their connection to Country.

Figure 2: Cyclists Riding Past Uluru
(*https://bicyclenetwork.com.au/newsroom/2023/05/10/eight-days-across-the-outback/*)

Table of Contents

1.0 Introduction & Objectives

The purpose of this project is to design, analyse, manufacture, test and validate a mobile phone mount with a linear quick release system for a motorcycle and bicycles. The design will fit all modern smartphones (excluding flip and fold phones), and have the ability to be used on the circular handlebars of the vast majority of motorcycles and bicycles. Whilst off the shelf fasteners can be used during assembly, other products such as magnets and metallic springs cannot be used. The design must allow for angular displacement in all axes (pitch, roll, yaw), and also be able to be orientated horizontally and vertically. Linear translation is optional. While the majority of real world designs utilise a rotational locking system to hold the phone in the holder, this new design must make use of a linear locking mechanism.

The phone attachment system must fit a phone of 150mm x 70mm x 10mm. However, the actual attachment itself must be contained with a 70mm x 70mm x 20mm box.

The mechanism to connect the bar side and the phone side must fit within 90mm x 90mm x 40mm. The system must be able to withstand an array of load cases, including acceleration/inertial, bump, user applied, and aerodynamic forces.

Figure 3: A Phone Mounted on Bike Handlebars
(https://www.decathlon.com.au/p/cycling-smartphone-mount-metal-decathlon-8587962.html)

2.0 Project Functions

The following figure shows a flow diagram of all the required (and optional) functions of the device/system, and how they are related. These are then taken and used in the following sections of the report, to idealise alternative concepts of the assembly and find appropriate solutions that satisfy all specifications.

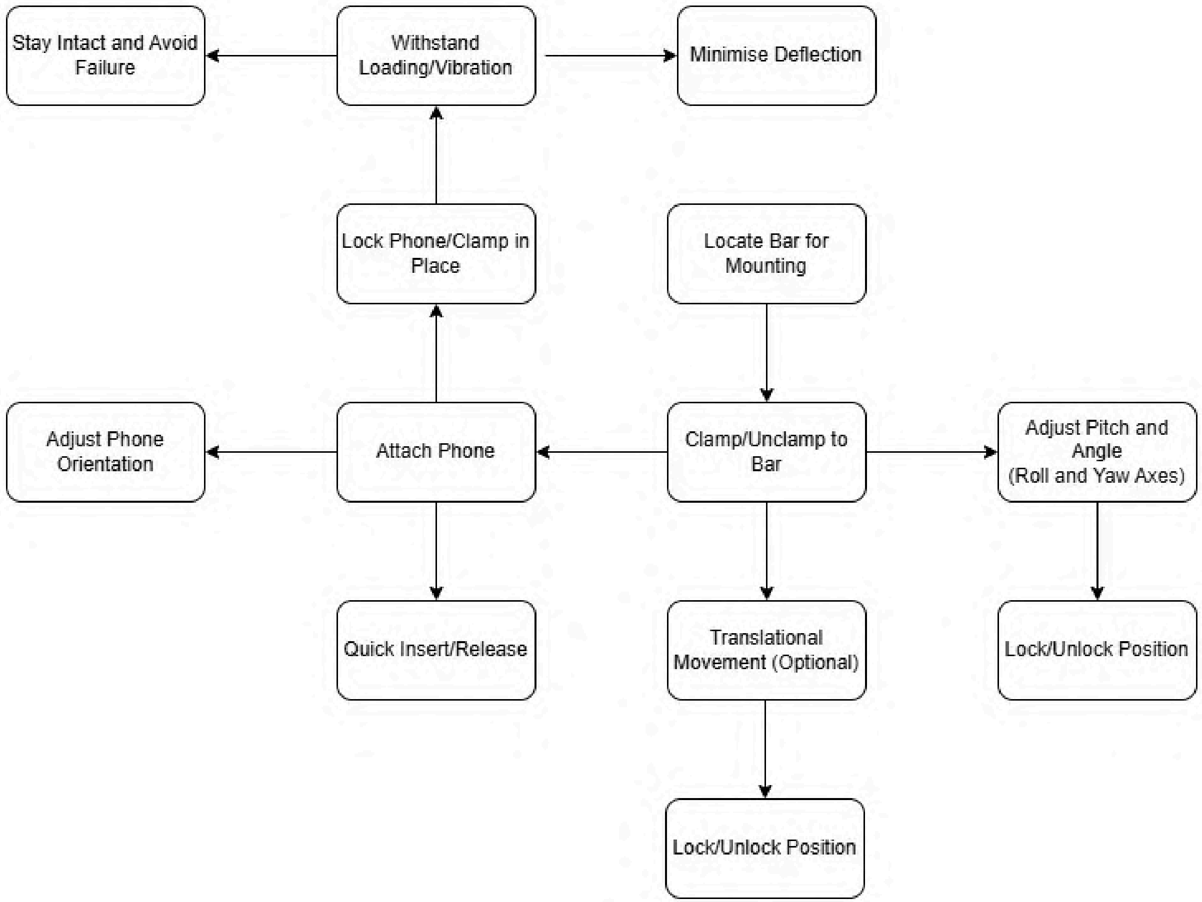

Figure 4: Functional Diagram

3.0 Project Factors

Human Factor

Rider: The device needs to work for both bicycles and motorbikes. Since motorcyclists generally wear thicker protective gloves this reduces the user's dexterity, meaning that the interface must be large and easy to use.

Ergonomics: The force required to operate the mechanisms should be comfortable for an average adult, have a quick release design and be intuitive.

Design Team: Five mechanical engineering students have been tasked with producing this product, who have varying backgrounds and skills with CAE software such as SolidWorks and Ansys, that will ideally compliment each other.

Money

Production costs: The clientele is a manufacturer that produces plastic goods in a high volume, using a new technology. This reduces the need for a large inventory of expensive plastic injection-moulding tooling for each model, hence reducing production costs.

Profit: The company ultimately needs to make an appropriate level of profit for the project to be viable. The product should ideally be cheap to manufacture and not use unnecessary material and/or production time wasting resources.

Machines

Fuse1 3D Printers: The design will feature parts made from Nylon 12 printed using the Fuse1 printers, since this is a requirement for all design groups there will be high demand for using the printers. This could cause congestion and wait times for printer access, particularly later in the design stages (semester) when the designs are more finalised and all groups are wanting to produce their final design.

Prusa 3D Printers: Most components of the design will be printed using PETG from these machines, which also reduces the demand for the Fuse 1 printers. Rapid prototyping of parts in this way can also speed up the overall process of testing design changes and part modifications.

Methods

Computer Aid: The model is to be designed in SolidWorks with parameterised features allowing customisation across things such as handlebar diameter. The design is to be tested using Ansys FEA to justify it being fit for use.

Testing: The expected loading and use cases will be practically simulated on a 1 inch set diameter handlebar, using weights hung down on a string. These weights will be used to depict the forces experienced under standard operating conditions aboard a motorbike, assuming the relatively static testing conditions can replicate dynamic conditions at a higher loading, i.e., 300 g at 3g (dynamic) = 900 g (static).

Operation: The design will feature a linear attachment motion of the phone to the mount, with available adjustment in roll, pitch and yaw with secure locking mechanisms. A compliant mechanism will also be incorporated, acting as the quick release feature for the phone attachment.

Minutes

Timeline: The deadline for the design of the product is clear, being due in Week 11. Preliminary designs are due at an earlier date, in Week 5. This ensures plans are on track and encourages consistent investment of work and effort along the way.

Scheduling: The team will meet regularly in and outside of class with minutes being taken to record what is discussed/worked on to ensure that the timeline and deadlines are upheld.

Materials

Nylon 12: The material required for the phone attachment device in the final product. A synthetic engineering thermoplastic known for good dimensional stability, low moisture absorption, and excellent fatigue resistance.

PETG: This is the material used for the majority of the design (excluding the phone attachment device) Preliminary designs and prototyping of the phone attachment device are to be printed in PETG to save time and costs rather than printing in Nylon 12.

Fasteners: General hardware (nuts, bolts) can be used in the parts attaching to the bike.

4.0 Requirements and Specifications

Requirement	Criteria	Weighting	Specification
Strong Grip to Handlebar	Pull off force in N	10	> 40 N
Strong Grip to Phone	Pull off force in N	10	> 30 N
Adjustability of Phone’s Position	Degrees of Freedom of Phone Positioning	7	> 3 DOF (Rotationtal)
Adjustability of Phone’s Orientation	Degrees of Adjustability of Phone Rotation	7	360°
Applicability to Different Phones	Size of Phone Attachment	2	< 70 x 70 x 20 mm
Adjustability to Different Handlebars	Range of Handlebar Diameters	5	20 - 40 mm
Deflection During Riding	Millimeters of Deflection During Riding	6	2 mm
Ease of Use	Peak User Force in N	8	< 5 N
Durability	Number of Cycles Before Fatigue Failure	6	1,000
Low Weight	Mass of Entire Mount	2	< 200 g
Compact Size	Overall Dimensions of Mount	N/A	< 150 x 70 x 10mm
Aesthetic	Judged Qualitatively	1	N/A
Risk of User Injury	Corner Radii in Millimeters	N/A	>= 0.5 mm

Table 5: Requirements and Specifications

5.0 Creative Solution Generation

5.1 Morphology Table:

The following table outlines the key functions that the design needs to achieve and potential options/alternatives, within the scope of the project, that may be plausible to finding an appropriate solution to the problem for the entire part/assembly.

	Options/Alternatives			
Functions	**A**	**B**	**C**	**D**
Attach to Bar	Re-Usable Adhesive	Rope/Pulley System	Press Fit onto Bar (Held with Friction)	Bolted Members
Adjust Pitch/Angle	Gears/Notches or Incremented Modules/Steps	Arm with Pivot Point/s	Ball Joint/s	
Lock/Unlock Position	Quick Release Lever	Threaded Fastener/Member over Ball Joint	Other Bolted Member System	
Locate Phone	Input Shaft into Hub	Buckle	Locating Boss	
Lock Phone	Quick Release Lever Clamp	Compliant Push Latch	Spring Loaded Push Button	Buckle

Table 6: Morphology Table

5.2 Attach to Bar

The aim of the attachment to the bar is to provide a rigid configuration that enables the mount to hold the entire assembly attached to it in place and prevent failure during use, i.e., phone mount falling off the bike while riding. Once in position, it is unlikely that the user will move/release the location of the mount regularly, however if changing between bikes and/or users, this still could occur and needs to be considered appropriately.

5.2.1 Re-Usable Adhesive

The potential options for a re-usable adhesive include:

- Apply glue every time
- Adhesive and/or tacky strip
- Double-sided tape

Conceptually, this idea is unfavourable particularly for serviceability and strength, as any form of re-usable adhesive is unlikely to withstand the loads and vibrations experienced during operation (permanent glue bonds may possess the required strength, in theory, but are then not able to be removed easily). The state of the bike's handlebars are also unlikely to be suitable for regular adhesion, due to their exposure to the environment and external factors, leading to poor surface conditions and inadequate bonding.

Figure 7: Handlebar Adhesive Mounting
(https://www.bike-components.de/blog/schrauben/4/)

5.2.2 Rope/Pulley System

Using a pulley or rope to fasten the mount to the handlebar is a good alternative to fasteners. The installation process is both toolless and efficient. Using a rubber strap would also help decrease weight, and reduce risks of scratching paint on the handlebars. However, this rubber strap will likely not be as rigid as a bolted mounting solution, as the rubber clamp will be able to deform and vibrate under the loadings it will face.

Figure 8: Rubber Strap Handlebar Mount
(https://www.amazon.com.au/Lsan-Handlebar-Mount-Easy-Install/dp/B09XHX2K98)

5.2.3 Bolted Members

The alternative methods of fasting the members of the mount down to the handlebar include:

- Singled bolted hinge
- Multiple fastener clamping method

Using a method that involves fasteners is the most simple solution and is common for other handlebar mounts on the market. This comes from the fact that it provides a capable and rigid mounting method with little complexity of installation for most users. However, it is slightly more tedious and requires more steps than potential alternatives.

Figure 9: Bolted Clamping Method

(https://www.amazon.com.au/Ataputa-bicycle-Aluminum-Standard-Handlebar/dp/B00FS6RHB6)

Due to these factors, this option is deemed the most feasible and reliable. It provides the required strength and rigidity, and can be designed to be relatively simple to install and service. The confidence in reliability and resistance to vibration makes it the appropriate solution for the intended application.

5.3 Adjust Pitch/Angle

The phone mounting system requires a level of adjustment for the user to be able to modify the positioning of their device to suit their sight and/or comfort. This function is contingent on practicality and functionality, as well as ensuring that the integrated design does not become too complex or threaten manufacturability.

5.3.1 Gears/Notches

An idea for the adjustment of pitch/angle is to use a gear or notching system that allows the movement of the mount system (or a potential arm) in set steps or increments, in each direction. This was inspired by the shifting/clicking of gears into place from a curved shifter design seen in some car models. However, this only allows specific adjustments in a limited step size, restricting full tuning capabilities and/or precision of phone mounting locations. There is potentially also risk of jamming or binding of gears and/or other components.

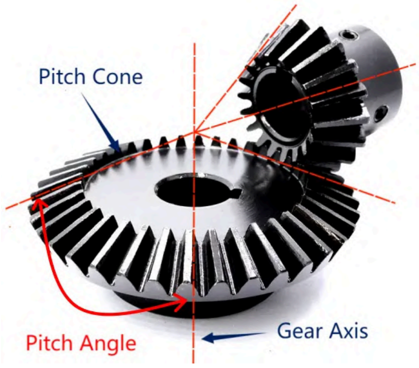

Figure 10: Bevel Gear Diagram

(https://www.chamolgear.com/bevel-gear-pitch-angle/)

5.3.2 Ball Joint

The most widely used answer for assemblies of this description utilise a ball joint of some variation to adjust the angle of the mount, as it is quite simple and very intuitive for user adjustment. The geometry is also relatively easy to manufacture, particularly in comparison. A ball joint also has the least limitations and most degrees of freedom.

While there is the potential option for multiple ball joints, the additional pitch/angle adjustment is unlikely to be necessary, and would merely result in greater manufacturing and/or assembly burden. In a similar sense, a pivot point design mechanism would be more restricted in its application. Hence, for these reasons, a singular ball joint will be chosen.

Figure 11: Ball Jointed Mount
(https://www.rubbermonkey.com.au/GoPro-Magnetic-Latch-Ball-Joint-Mount-for-HERO13-Black)

5.4 Lock/Unlock Position

In relation to its adjustability, the positioning of the phone must also be fixed in position during use, in order to withstand the required vibrations and loading, which would otherwise compromise the system and likely lead to sudden failure.

5.4.1 Quick Release Lever

A solution that could work with any pitch/angle adjustment is the use of a lever that locks into place in order to hold, clamp or anchor the placement of the phone mount assembly. For example, in a gear/notching system, it would require a lever of some form to select between positioning, or with a ball joint, the locking/unlocking mechanism could be configured in order to push a device down to clamp on the ball. This would be able to provide serviceable, quick adjustments, however there is concern that the lever needs to be small enough to remain within constraints and avoid obstructions, which may affect its usefulness in this regard.

5.4.2 Threaded Fastener/Member over Ball Joint

The most simple and intuitive method to lock a pivot point or ball joint would be to use a threaded fastener or member that locates the joint and locks it into place. For example, having a nut over a ball joint which clamps the ball down into a socket. This will require the use of a snap fit ball joint to assemble the pieces together.

Due to the chosen selection of a ball joint, and the constraints of the project, a design of the clamping nut mechanism to lock/unlock the positioning has been chosen.

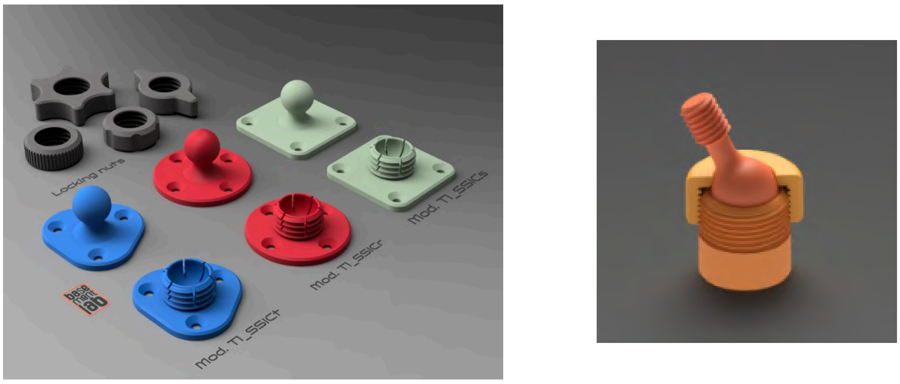

Figure 12: Snap Fit Locking Nut Ball Joint
(https://www.printables.com/model/1690221-snap-fit-ball-joint-with-locking-nut-giunto-sferic)

5.5 Locate Phone

Assembly of the phone onto the holder should be easy to locate and arrange consistently. A potentially optimised design may also blend this function with the following, of locking the phone onto the mount itself.

5.5.1 Input Shaft into Hub (or Locating Boss)

The arrangement of an input shaft (male) that sets in place into a hub (female) of some description would be a plausible option for locating the phone onto the remaining assembly pieces. This could also be done with a “locating boss”, which is in essence the same idea, merely executed in an alternative manner (likely unfeasible without the use of magnets). Implementing this would create a rigid, linear connection (rather than rotational, as per the project constraints), however would increase excess outwards stickout from the mount, increasing the moment arm of the connection and moving the phone closer to the rider, which may cause discomfort for the positioning of the rider and/or their phone view.

Figure 13: Locating Boss
(https://www.peakdesign.com/en-au/products/universal-bar-mount)

5.5.2 Buckle

Similar to common everyday items, such as bags and certain seatbelts, a quick release buckle may also be incorporated for this function. The design of a buckle naturally has a locating/locking mechanism built in, and is intuitive for the purpose of the user. Due to this, and its relatively slim design to fit within the project constraints, a buckle was designated as the preferred method.

Figure 14: Quick Release Buckle
(https://www.amazon.com.au/Baitoo-Military-Adjustable-Replacement-Backpack/dp/B0953NKTVL)

5.6 Lock Phone

Once the phone is located onto the mount assembly, it must also be locked into position in order to prevent movement and ensure its rigidity during use. The main purpose here being that it must be ensured that the phone does not fall off from its stand at any point, which could damage components and/or the expensive phone itself.

5.6.1 Quick Release Lever Clamp

A system that is able to clamp onto the phone, and then also include a user-friendly quick release locking lever, would allow for both the locking and unlocking of the phone onto the mount, within essentially the same motion. However, this mechanism would likely only be feasible on a shaft and hub locating design, which was decided against.

Figure 15: Quick-Release Lever Clamp
(https://www.1080riders.com/products/a-kick-quick-release-lever)

5.6.2 Buckle

Given proven history of the ability for the design of a buckle to both locate and lock the phone position into place, with relative simplicity, easy manufacturability, and low parts count, the option was nominated for the design of the phone mount.

6.0 Preliminary CAD Design/s

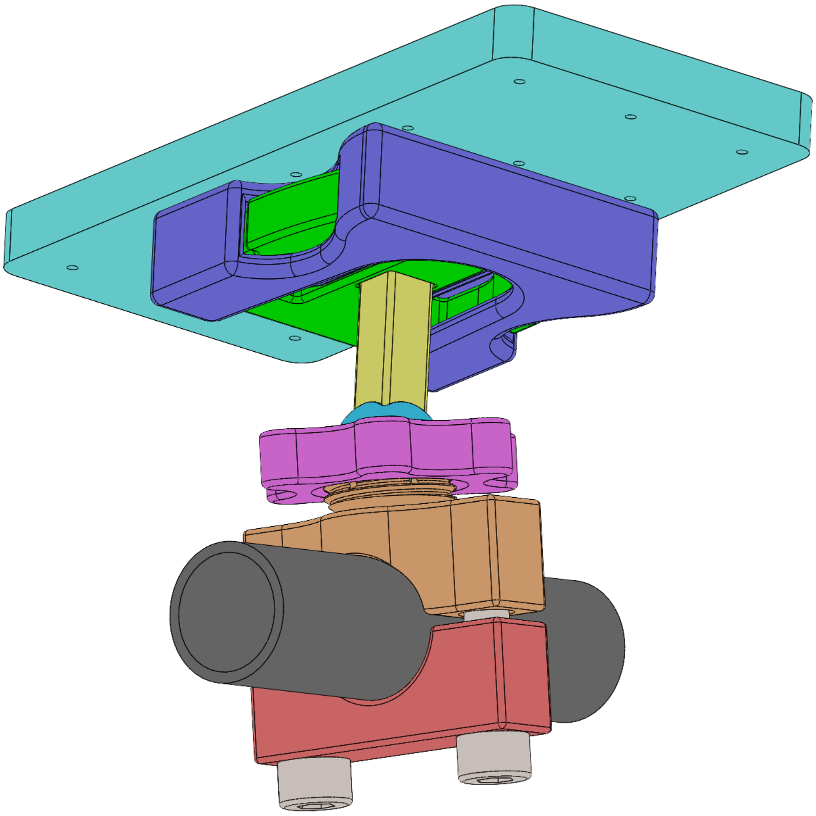

Figure 16: Preliminary CAD Model

Figure 16 showcases the preliminary CAD model of the designed phone mount assembly. Selections of component design options were predicated upon the analysis and relative scores conducted in the morphology table and creative solution generation section of the report.

Feature (1) in Table 17 showcases the buckle system design utilised. To allow for maximum adhesion, the surface area on the top of the negative buckle component has been enlarged near to the maximum boundary limits. There are centering vanes located inside the same part, along with the tapered central arm of the positive piece, which allows for consistent and easy insertion. The design of the main compliant arms have been iterated using rapid prototyping of 3D printing, in order to maximise ergonomics and further aid user friendly compatibility.

The positive buckle piece is fastened into a tapped hole in the main pillar (see Figure 1 of Table 18), as is the ball. During testing, it was found that printing these as one piece resulted in the shearing of the component at very low stresses, particularly at the edge between pillar and buckle. The combination of these into a single unit, violated the constrained boundary box. Hence, making them separate pieces was necessary.

A square shaped pillar was implemented to prohibit the buckle and the ball from twisting relative to each other, which would introduce unwanted movement when the system was locked in place.

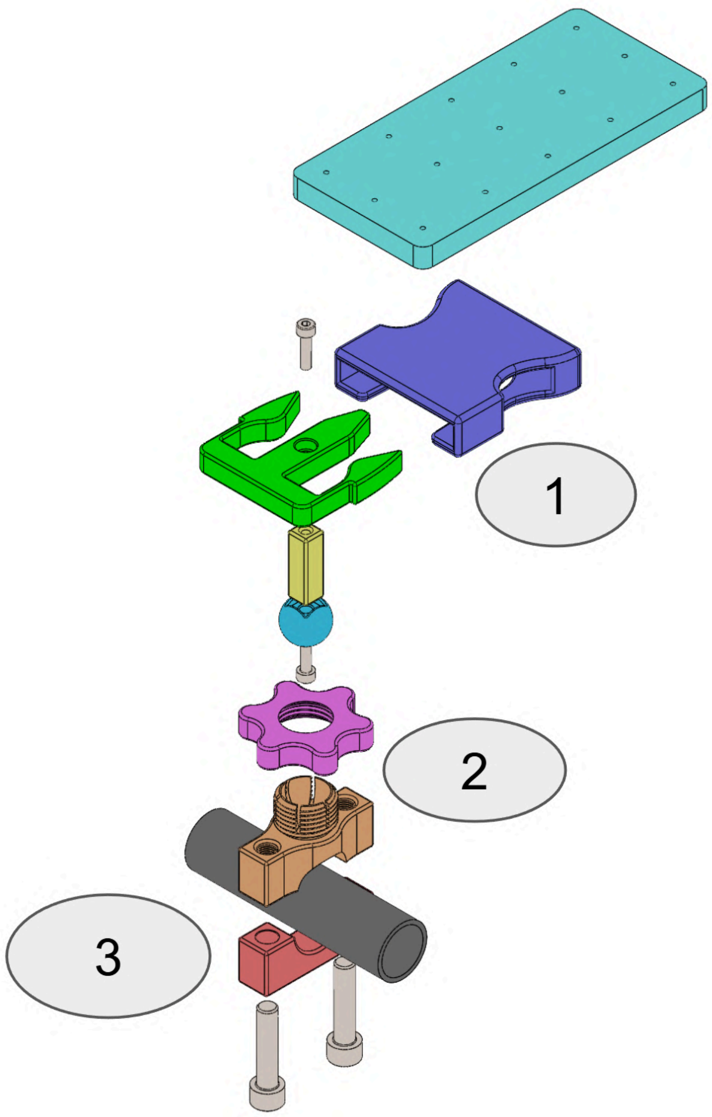	**1**	Positive (green) and negative (dark blue) buckle pieces. Used for both locating and locking the phone.
	2	Ball (light blue), clamping nut (purple), and threaded socket (orange). Used to provide angular and rotational displacement of the phone, before tightening and locking into position.
	3	Upper mount (orange) and lower mount (red). Used to clamp onto the handlebar. The upper mount also features the threaded ball socket.

Table 17: Key Features of the Phone Mount

Feature (2) is responsible for permitting angular and rotational freedom, while also being able to lock the ball joint in place. The upper handlebar mount features a negative ball socket with a thread on the exterior of the boss extrude. The clamping nut features an opposing internal thread, along with an internal chamfer. As the nut is tightened onto the socket, this chamfer presses against a matching chamfer. Six slots cut in allow the socket to flex inwards under pressure, clamping down onto the ball joint to lock the assembly in place.

Feature (3) is a relatively simple mount at this stage. The lower mount features clearance holes which allow the bolts to pass through into the tapped threads of the upper mounts, in order to clamp the two halves together onto the handlebar (see Figure 2 of Table 18). The mount incorporates excess material surrounding the bolted regions for added stiffness, which minimises local deflection within the mount itself. Along with a vertical offset from the handlebar, this results in a tight clamp on the bar. This is key to preventing the unit from spinning around the handlebar under loading conditions.

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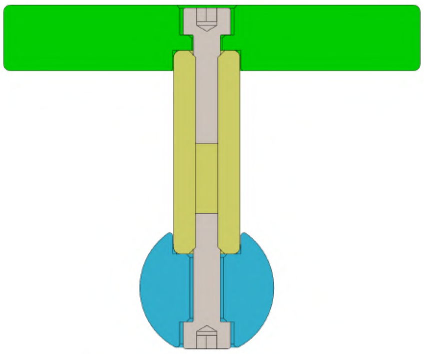
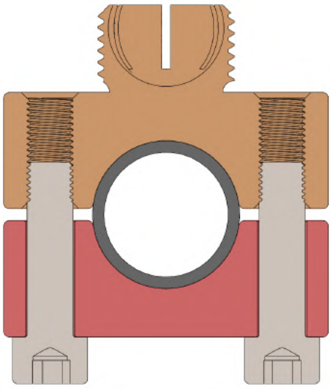
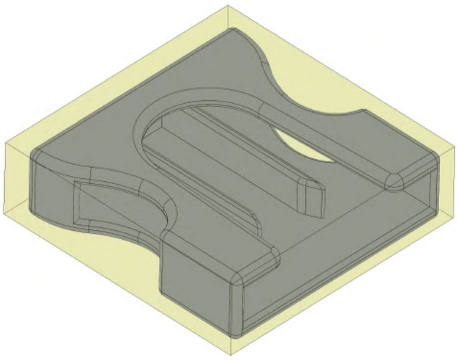
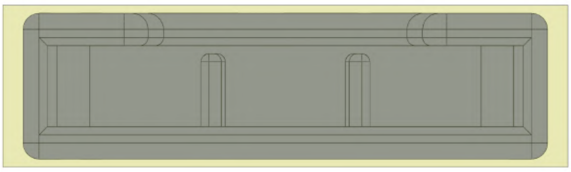
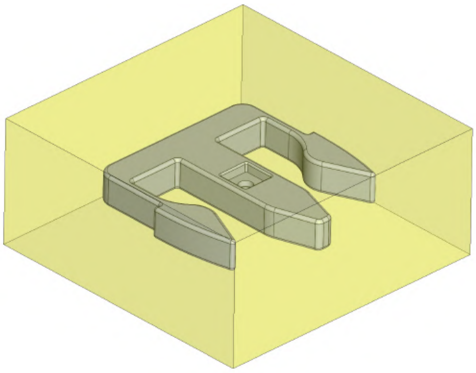
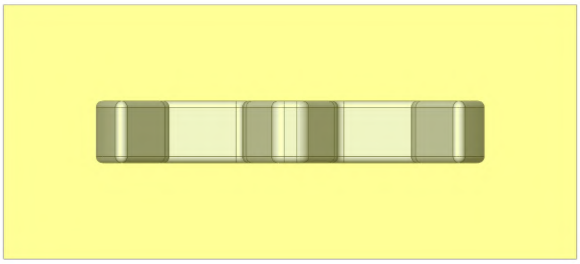
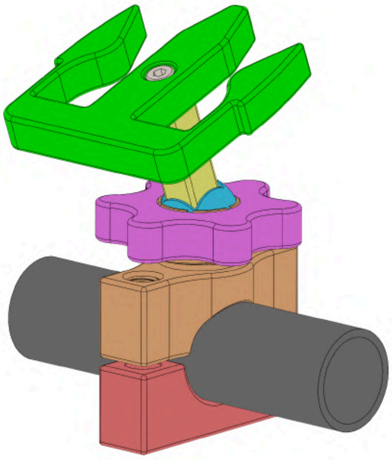
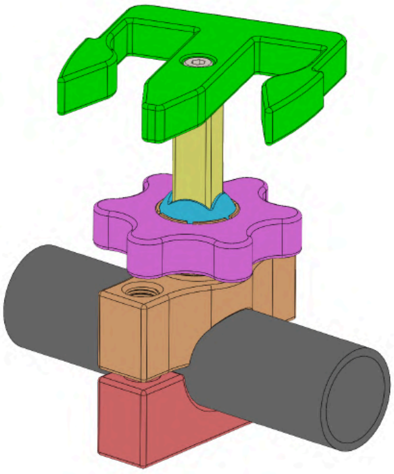
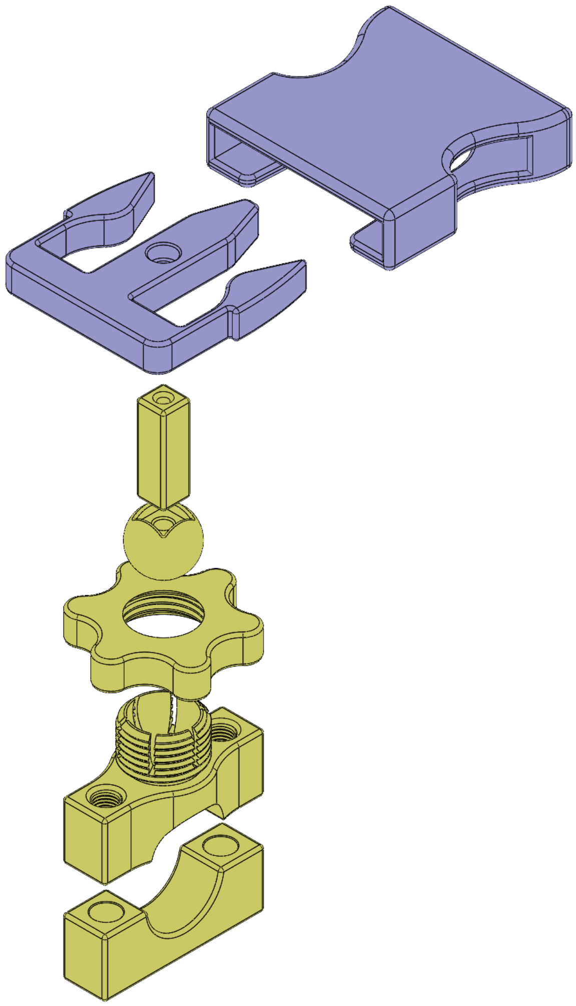
1		2	
3		4	
5		6	
7		8	

Table 18: Preliminary CAD Breakdown

As can be seen in Figures 3-4 and 5-6 of Table 17, both sides of the phone attachment mechanism (in the case of this design, the buckle), fit within their assigned limits. While there is room to play with on the positive side of the buckle, the current design has been proven to work reliably through prototyping and testing. This also gives room for future alterations if necessary. As an example, if a change of material resulted in higher stress or strain values, the geometry of the arms could be modified to suit.

Figures 7 and 8 of Table 17 demonstrate how the ball and socket unit allows for a complete 360 degrees of yaw rotation, along with sufficient pitch and roll adjustment. This is necessary to allow for fine tuning adjustment of the unit to account for a wide range of motorbike and driver combinations.



Colour:	Material:
Purple	Nylon 12
Yellow	PETG

Figure 19: Breakdown of Materials

The components of the buckle are to be produced from Nylon 12, being a requirement of the project. However, the rest of the assembly is to be produced from PETG, due to being readily available, inexpensive, and sufficiently strong. FEA analysis later detailed in this report is conducted with all components modelled as Nylon 12, due to their similarity in key material properties (i.e., Young’s Modulus). These photos can be found in Appendix A along with more extensive angles and split views.

7.0 CAE Analysis

7.1 Objective of the Analysis

The mount transfers the applied load from the mobile phone attached to the motorcycle/bike handlebar through the following series load path: female buckle → male buckle → pillar → ball joint → handlebar clamp → handlebar tube. Due to this, the applied loading is transferred through each component and interface. Therefore, the overall structural performance of the assembly is dependent on the strength and stiffness of each component/connection, and will likely fail at the weakest interaction point.

Particular focus will likely be on the buckle connection and ball/socket joint, where the interfaces rely on friction and elastic preload to resist relative movement, rather than a fully positive mechanical interlock.

The client's requirements specify that the phone mount must maintain structural integrity, retain its angular and positional adjustment, limit movement/deflection of the phone and minimise vibration under expected user applied and/or aerodynamic loading. The purpose of the CAE analysis is to assess these requirements during the concept development stage, in case adjustments were required.

The investigation primary focuses on the following structural requirements:

- **Maximum Stress** - ensuring the peak stress within components remains below the allowable material stress.
- **Deformation** - understanding the maximum deflection result, and confirming if the stiffness requirement keeps the outcome below 2 mm

7.2 Applied Loading & Boundary Conditions

The physical mounting arrangement was represented in ANSYS through the following boundary conditions and loading:

- The handlebar connection between the mount and motorcycle/bike was modelled as a **fixed support**.
- The 2x M8 handlebar mount clamping bolts were represented using spring elements (preload = 10,000 N, stiffness = 331,340 N/mm).
- Cylindrical joints were applied to the relevant bolt holes.
- Global accelerations were applied to the assembly to represent the expected acceleration and/or bump loading.
- A **vertical acceleration of 3g** was used to represent the major bump loading.
- **Longitudinal and lateral accelerations of 1g** were combined with the vertical loading to represent acceleration, braking and cornering events (based on the expected mechanical grip limits of the motorcycle/bike).
- The resulting loading conditions were also rotated to account for different orientations of the mount.

Refer to Appendix B. These acceleration values were parameterised within ANSYS and combined with the different mount and phone orientations, by splitting the vector components, to provide a total of 48 analyses cases.

7.3 Modelling Assumptions

Several assumptions were implemented in order to establish appropriate modelling constraints and provide a well defined analysis:

- The phone mount is not intended to remain intact and be structurally viable to withstand crash and/or impact loading. Hence, this analysis was excluded.
- The ball joint is assumed to become rigid once clamped, with sufficient pressure provided by the clamping nut to prevent slipping.
- The female buckle is assumed to be bonded to the phone, with the load transferred uniformly across the bond area.
- The phone is modelled as an aluminium block, assigned with relative density to represent the mass of a typical device.
- The material behaviour of the Nylon 12 material is assumed to be isotropic and homogenous.

7.4 Material Assignments

The material properties of Nylon 12 were assigned to all assembly components in the analysis, based on the following data:

Property:	Value:	Units:
Density	1000	kgm^-3
Young's Modulus	1.7E+09	Pa
Poisson's Ratio	0.4	-
Tensile Yield Strength	2.5E+07	Pa

Table 20: Engineering Data of Nylon12

(https://www.xometry.com/resources/materials/nylon-12/)

7.5 Finite Element Model

Threaded features of the handlebar clamp and ball joint clamp were suppressed and replaced with cylindrical geometries to reduce mesh complexity and computational time. Similarly, bolts were not explicitly modelled and were instead represented using spring elements and cylindrical constraints.

Angular parameters were created for the pitch and roll directions of the ball joint components. These parameters allowed the structural response of the mount to be evaluated at different phone orientations, including: 0°, 45° and 90°. Due to the approximate symmetry of the mount about two planes, a reduced set of representative angular configurations was analysed:

Case:	Pitch Angle:	Roll Angle:
1	0°	0°
2	29°	0°
3	0°	29°
4	16.5°	16.5°

Table 21: Pitch and Roll Angle FEA Positions

With these cases, and the different evaluations of phone orientation, this resulted in a total of 16 cases analysed (Refer to Appendix B).

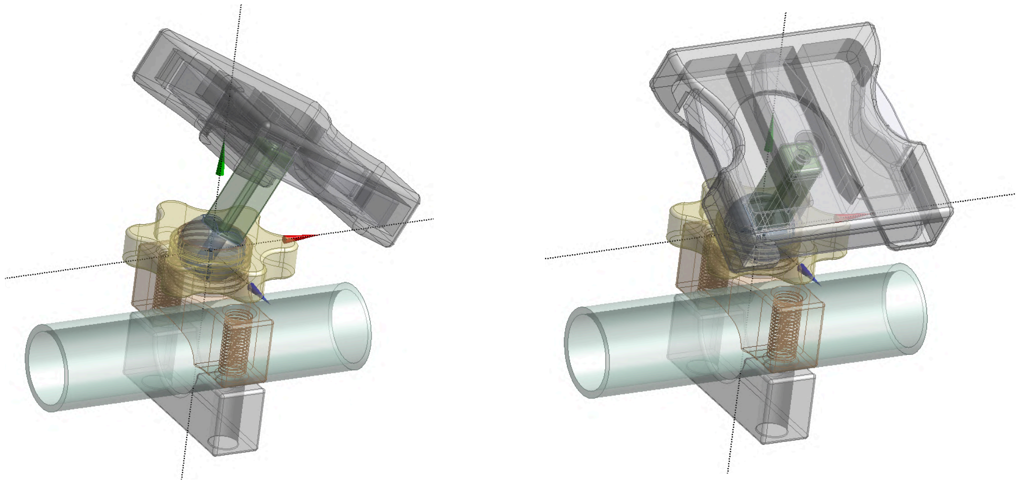


Figure 22: Ball Joint at 29° Roll (Left) and 29° Pitch (Right)

7.5.1 Contacts

Frictional contacts were applied between the handlebar and the mount to represent the clamping interface. A friction coefficient of 0.3 was applied on the Nylon 12 to steel contact surface, and to determine whether the clamping force from the 2x M8 bolts was sufficient to restrain the mount under the expected loading conditions. The male and female buckle interfaces were also modelled using frictional contact, with a coefficient of 0.15 to represent the Nylon to Nylon contact.

The remaining interfaces were bonded contacts in this preliminary stage of development. This includes: the pillar to the ball and make buckle, the clamping nut on the ball joint socket, as well as the ball joint itself (assuming that the clamping force prevents relative movement when tightened).

7.5.2 Mesh

A global element size of 2 mm was applied to all components in the mount assembly. This provided sufficient mesh refinement to capture the relevant geometric features and correlated stress concentrations. To reduce computational time, the handlebar was assigned a 5 mm mesh, particularly as it was not of primary importance or concern for this project.

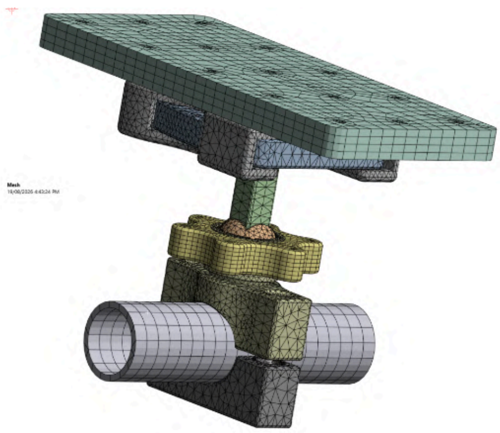


Figure 23: FEA Mesh of Assembly

7.6 FEA Results

The parameterised FEA results were compiled into an FEA tracking spreadsheet, refer to Appendix B. This allowed the stress and deformation results from each loading and orientation case to be compared.

As a result, the maximum stress across all cases was seen to be ~9 MPa. This is significantly below the report yield strength of 45 MPa, of Nylon 12, signifying a healthy factor of safety as currently constructed.

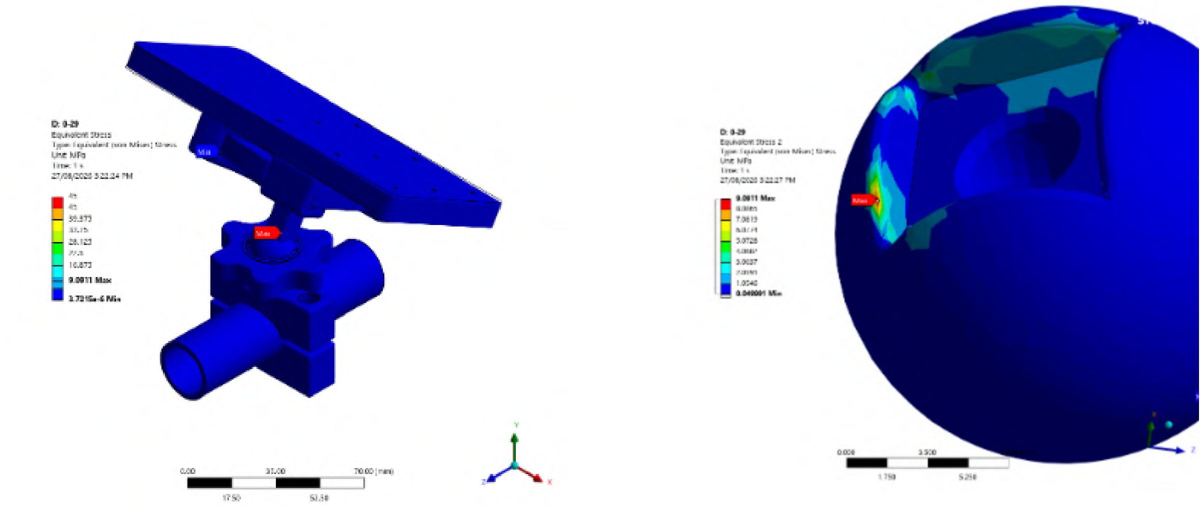


Figure 24: Maximum Stress Results, with Close Up View of the Ball

For the critical load case, the maximum stress occurs on the interface of the ball with the pillar. The edge of the ball at this connection experiences a local stress concentration.

In the majority of remaining load cases, the peak stress occurs around the handlebar clamping bolt interface, due to the applied force from the fastener/s. Despite this, the remainder of the assembly experiences relatively low stress.

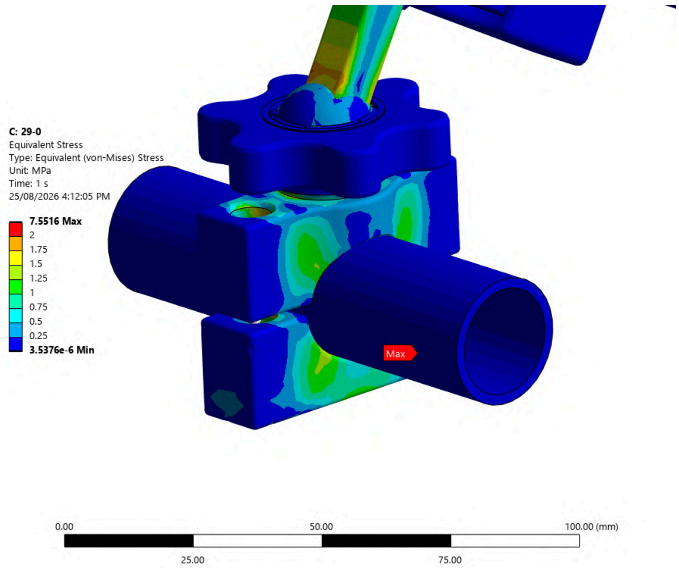


Figure 25: Contour Plot with Maximum Stress Scale Adjusted to 2 MPa

Across the cases included in the analysis, the maximum deflection was located on the phone model, where it was seen to be ~1.12 mm. This can be attributed to the gap between the female and male buckle components.

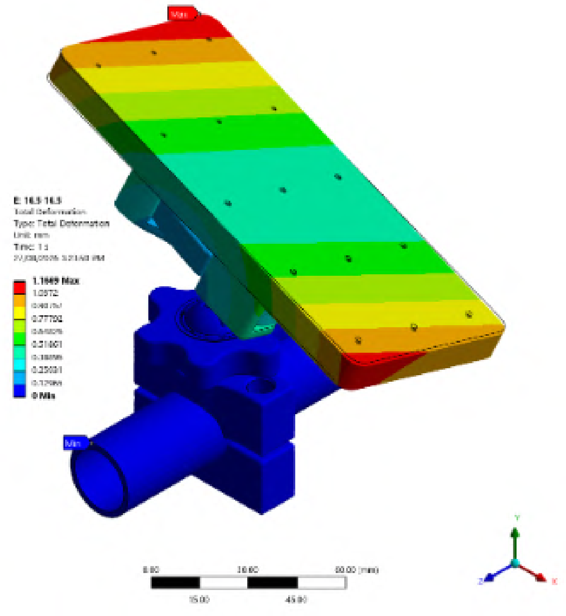


Figure 26: Contour Plot of Maximum Deflection

This maximum deflection is below the design requirement of 2 mm, indicating that the mount provides sufficient stiffness under the loading conditions.

The maximum strain within the mount components was hence calculated to 0.66%. This remains significantly below the reported maximum elongation of Nylon 12 at 10%, signifying that the components, particularly our compliant buckle mechanism, remains within the expected elastic deformation range.

Parameter:	**Design Requirement:**	**Maximum Result:**
Maximum Stress	45 MPa	9.091 MPa
Maximum Deflection	2 mm	1.1669 mm
Maximum Strain	10%	0.66%

Table 27: FEA Results

Overall, the FEA analysis demonstrated that the phone mount satisfies the primary structural and stiffness requirements under the expected acceleration and loading conditions.

7.6.1 Aerodynamic Loading

Upon further consideration, it was determined that an estimation of the Aerodynamic loading on the phone could be included in the FEA analysis. Assuming that the phone could be modelled as a flat plate, an assessment of the Coefficient of Drag was found and utilised in the following calculations of the drag force (Refer to Appendix B).

$$F_D = C_D A \frac{\rho V^2}{2}$$

where
F_D is the drag force
C_D is the drag coefficient
A is the reference area
ρ is the density of the fluid
V is the flow velocity relative to the object

$$C_D = 1.98$$

$$A = 0.15m * 0.075m = 0.01125m^2$$

$$\rho = 1.225 \frac{kg}{m^3}$$

$$V = 200 \frac{km}{h} = 55.56 \frac{m}{s}$$

$$F_D = 1.98 * 0.01125 * \frac{1.225 * 55.56^2}{2} = 42.116N$$

This analysis was undertaken with the phone at 45° angle, and 29° of pitch, as this puts the phone essentially normal with the direction of airflow. From this additional loading, the FEA results showcased a slight increase in stress and deflection. However, it remained essentially unchanged in every other manner.

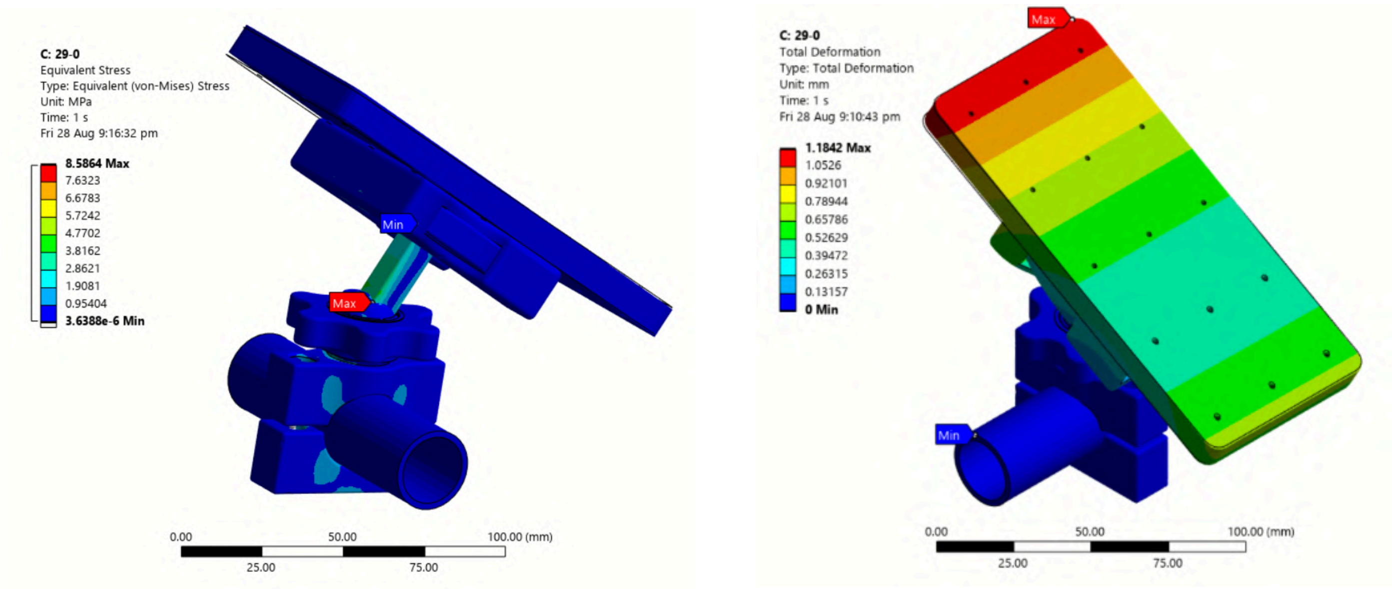


Figure 28: Stress and Deflection Contours with Aerodynamic Loading

Parameter:	**Design Requirement:**	**Maximum Result:**
Maximum Stress	45 MPa	8.5864 MPa
Maximum Deflection	2 mm	1.1842 mm

Table 29: FEA Results with Aerodynamic Loading

8.0 Prototype Manufacturing and Experimental Testing

To evaluate the physical feasibility of the assembly design, iterative prototyping was completed. Certain components needed to be tested in isolation and were printed individually to identify key issues with the geometry, tolerancing, or any compliance. These issues would be improved upon and consequently eliminated in future versions. A full rundown of every version of the model, along with an overview of the changes and driving factors behind such modifications can be found in Appendix C.

Figure 30: Preliminary Prototype

Filament	PLA or PETG
Infill	10% ~ 15%
Layer Height	0.15 mm

Table 31: Standard Print Settings for Prototyping

The print settings were held consistent for the majority of the prototyping period, given there no issues were found in relation to them. The filaments used were either PLA or PETG, usually depending on what was readily available. The exception to this was the positive buckle piece.

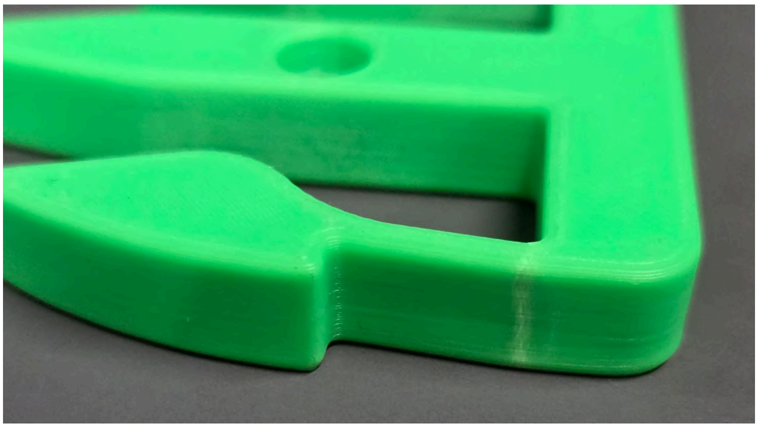


Figure 32: Fatigue Observed in PLA Positive Buckle Prototype

Vigorous detent testing was conducted by inserting and removing the positive buckle from the negative housing. The PLA positive buckle quickly showed signs of fatigue, as visible in Figure 32 above, of the white lines caused by stress whitening. In order to solve this, the piece was instead printed out of PETG (as seen in Figure 30), given its less brittle nature, which allows for more deflection before yield failure.

Physical testing was conducted by hanging weights from the buckle. The assembly was clamped to a 1 inch diameter steel tube (see Figure 33 below).

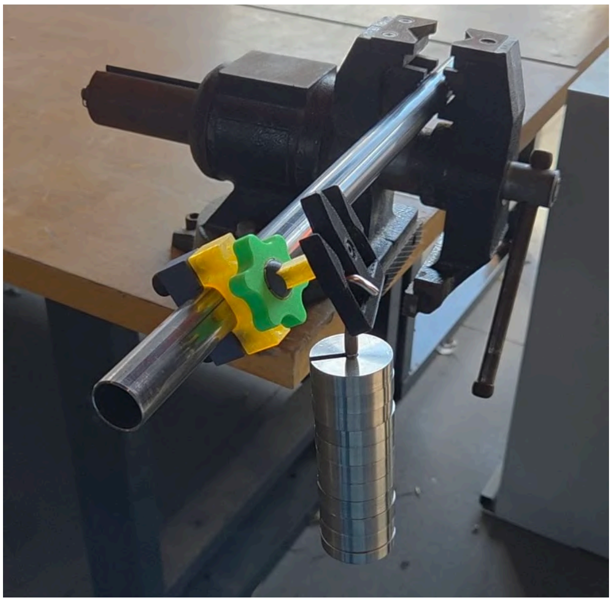


Figure 33: Initial Testing Setup

Initial testing showed the assembly held steady with 1200g of vertical load. However, when this load was attached via a string and abruptly dropped, the design appeared to lack sufficient friction to hold the ball, which caused slipping of the joint. A larger nut was manufactured (seen in Figure 34) to try and achieve more torque and thus more clamping force, however the problem persisted.

Figure 34: Testing Before Dropping 1200g, Figure 35: Testing After Dropping 1200g

It was determined that the 1200g static loading was equivalent to the worst load case, (~300g at 3g, which is roughly ~900g under static loading), and thus the slipping was not a critical issue. To try and determine if the assembly was close to failure, another test was conducted. This time, the unit was placed completely horizontally to induce the greatest moment, and the load was increased to 4000g.

Figure 36: Improved Testing Setup, Figure 37: Maximum Deflection on Dial Gauge

Under these new testing conditions, the system retained the ball joint to no slipping. A dial indicator was used on the top surface of the buckle, where the maximum deflection was measured to be 1.27 mm (refer to Table 38 for results and comparison to FEA). FEA analysis was conducted at 900g, and resulted in a maximum deflection at the buckle of ~0.3 mm. Under the assumption that deflection scales linearly with applied load, the comparison between physical testing and the scaled FEA results correspond reasonably well. See Table 38 for the breakdown.

Setup	Maximum Loading	Maximum Deflection
Physical Testing	4000g	1.27 mm
FEA	900g	~0.3 mm
FEA (Scaled)	4000g	~1.3 mm

Table 38: Physical Testing Results/Comparison at the Buckle

9.0 Design Summary and Next Steps

The project has progressed from the initial brainstorming, OFFERS analysis and creative solution generation to prototyping and simulation validation with a working model. In order to produce a design that fit together and met the required specifications, several iterations of prototypes were developed and iterated upon, in order to make improvements from the previous issues identified. The current design incorporates a buckle and ball joint which allows for easy and intuitive operation by the user. These specific design choices are also common features many people would be familiar with. Importantly the design maintains its structural integrity under loading. This was proven in both the physical testing and the FEA completed through Ansys.

However real world testing with a bump load of 3g (approximately 1.1kg) caused the ball to slip in the socket, although not to the maximum point which indicates the design is close to limited deflection. This is the key area for improvement as we work to increase the clamping force to keep the ball joint from moving. This will require further brainstorming and testing. Furthermore, the next step must also utilise the printing of Nylon 12. Since this is the required material for the final product, it is important to test the compliant mechanism in Nylon. In PLA the buckle prong deflection causes significant stress and fatigue, whereas the PETG prototype is far more flexible. The behaviour of Nylon 12 will likely vary from these and will take some adjustments and fine tuning to the CAD model/geometry, to ensure the buckle will provide adequate flexibility and reliability without drastic fatigue failure.

11.0 References

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Appendix

Appendix A: Preliminary CAD Design Screenshots

Prelim CAD Design Screenshots

This file features in depth screenshots of the preliminary CAD model in its current form. Further images not included in this document can be found here with split views and alternative angles.

Appendix B: FEA Tracking

Phone Mount FEA Tracking

The FEA Tracking spreadsheet documents each iteration of the ANSYS model. To imitate different user scenarios the mount was rotated varying degrees in roll, pitch and angle on the handlebars. All of the set up data such as forces, materials, contacts, etc are tabulated for each scenario, as are the results. There are accompanying images that show the setup with the cad model, and the results.

Appendix C: Version Tracking

Team 47 Version Tracking

Team 47 Version Tracking (Overview)

The Version Tracking spreadsheet served as a tool for us to log each iteration of the design. It features images of printed parts and CAD models. They focus on changes we made to the design based on prototyping. We recorded issues that were uncovered in each stage and the solutions applied.

The overview document provides the main issues found for each iteration, along with the modifications and their driving factors. It is a tidier version of the spreadsheet with more detailed explanations of the changes made in each iteration.